

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B. Tech (CBCS) Degree Examinations
 Summer Semester July 2024
EC2106-1 – SIGNALS AND SYSTEMS
 (EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

- If $x(t) = \begin{cases} 4 & 0 < t < 2 \\ 0 & \text{Otherwise} \end{cases}$, Find the energy of the signal $x(t)$.
 A) 32J B) 28J
 C) 34J D) 16J
- Find the fundamental period of the signal $x[n] = 5\cos\left(\frac{2\pi n}{3}\right)$
 A) 5 B) 3
 C) 6 D) 4
- The scaling property of an impulse function in continuous time domain is given by
 A) $\delta(at) = \frac{1}{2a}\delta(t)$ B) $\delta(at) = a\delta(t)$
 C) $\delta(at) = \frac{2}{a}\delta(t)$ D) $\delta(at) = \frac{1}{a}\delta(t)$
- If $x(t) = tu(t)$, then $x(t)$ is a
 A) Energy signal B) Power signal
 C) Neither Energy or Power signal D) Energy and Power signal
- $x[n] * \delta[n - n_0] =$
 A) $x[n - n_0]$ B) $x[n_0 - n]$
 C) $\delta[n - n_0]$ D) $\delta[n_0 - n]$
- $\delta[n - \alpha] * \delta[n - \beta] =$
 A) $\delta[n - \alpha]$ B) $\delta[n + \alpha + \beta]$
 C) $\delta[n - \beta]$ D) $\delta[n - \alpha - \beta]$
- A continuous time LTI system is memoryless if its impulse response is of the form
 A) $h(t) = c\delta(t)$ B) $h(t) = c - \delta(t)$
 C) $h(t) = c + \delta(t)$ D) $h(t) = c\delta(ct)$
- For a continuous time LTI system, the step response is the ----- of its impulse response
 A) Running summation B) Delayed version
 C) Running integral D) None of these
- The Time domain convolution property of Fourier series is given by
 A) $Z(k) = TX(k) * Y(k)$ B) $Z(k) = TX(k)Y(k)$
 C) $Z(k) = X(k)Y(k)$ D) $Z(k) = X(k) * Y(k)$
- How does Fourier series make it easier to represent periodic signals?
 A) Sinusoidally related B) Periodically related
 C) Harmonically related D) Exponentially related
- The Duality property of DTFS is given by
 A) $X(k) = \frac{1}{N}x(-k)$ B) $X(n) = \frac{1}{N}X(-k)$
 C) $x(n) = \frac{1}{N}x(-k)$ D) $X(n) = \frac{1}{N}x(-k)$

12. What are the two types of Fourier series?
 A) Trigonometric only
 B) Trigonometric and exponential
 C) Trigonometric and logarithmic
 D) Exponential and logarithmic
13. The frequency shift property of Fourier Transform is given by
 A) $e^{j\beta t} x(t) \leftrightarrow X(j(\omega - \beta))$
 B) $e^{-j\beta t} x(t) \leftrightarrow X(j(\omega - \beta))$
 C) $e^{j\beta t} x(t) \leftrightarrow X(j(\omega + \beta))$
 D) None of these
14. The Fourier transform of $x(t) = 1$ is given by
 A) $\pi\delta(\omega)$
 B) $\delta(\omega)$
 C) $2\pi\delta(\omega)$
 D) π
15. A signal $x(t) = \cos(5\pi t) + 0.5 \cos(10\pi t)$ is ideally sampled. The minimum sampling frequency is given by
 A) 10Hz
 B) 20Hz
 C) 30Hz
 D) 40Hz
16. Which of the following is the process of 'aliasing'?
 A) Peaks overlapping
 B) Phase overlapping
 C) Amplitude overlapping
 D) Spectral overlapping
17. The Z-transform of $x(n) = \delta(n - n_0)$ is
 A) Z^{-n_0}
 B) $\frac{1}{Z^{n_0}}$
 C) Z^{n_0}
 D) $\frac{1}{Z^{-n_0}}$
18. The following method is used for finding inverse Z transform
 A) Runge Kutta method
 B) Partial Fraction method
 C) Power transform method
 D) Differential method
19. If $x[n] = -u[-n - 1]$, then $X(z)$ is given by
 A) $\frac{z}{z - 1}$, ROC: $|z| > 1$
 B) $\frac{z}{z + 1}$, ROC: $|z| < 1$
 C) $\frac{z}{z - 1}$, ROC: $|z| < 1$
 D) $\frac{z}{z + 1}$, ROC: $|z| > 1$
20. If $x[n] = u[n - 2]$, then $X(z)$ is given by
 A) $\frac{z}{z + 1}$, ROC: $|z| < 1$
 B) $\frac{z^{-1}}{z - 1}$, ROC: $|z| > 1$
 C) $\frac{z^{-1}}{z - 1}$, ROC: $|z| < 1$
 D) $\frac{z}{z - 1}$, ROC: $|z| > 1$

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

	Marks	BT*	CO*	PO*
1. a) Classify the below signals into energy and power signals and evaluate its energy or power. i. $x[n] = \left(\frac{1}{2}\right)^n u[n - 2]$ ii. $x(t) = e^{-at} u(t)$	8	L*3	1	1
b) Determine whether the following signals are periodic or not, if periodic, find the fundamental period i. $x(t) = \cos t + \sin \sqrt{2}t$ ii. $x[n] = \cos\left(\frac{\pi}{4}n\right) + \sin\left(\frac{\pi}{3}n\right)$	8	L3	1	1
2. a) Convolute the two discrete sequences given below $x_1[n] = \alpha^n u[n]$ and $x_2[n] = \beta^n u[n]$	8	L3	2	1
b) Convolute the two continuous time functions given below $x_1(t) = u(t + 1)$ and $x_2(t) = u(t - 2)$	8	L3	2	1
3. a) Evaluate the step response for the following LTI system i. $h[n] = \delta[n] - \delta[n - 1]$ ii. $h(t) = u(t + 1) - u(t - 1)$	8	L3	2	1

- b) A discrete time LTI system is defined by the difference equation,
 $y[n] - 0.6y[n-1] = x[n]$, $x[n] = (0.4)^n u[n]$, $y[-1] = 10$. Find the total response of the system.

8 L3 2 1

Unit – II

4. a) Obtain the DTFS coefficients of $x[n] = \cos\left(\frac{6\pi}{13}n + \frac{\pi}{6}\right)$. Sketch its magnitude and phase spectra.
- b) Determine the FS representation for the signal $x(t) = \cos 4t + \sin 8t$ and draw its magnitude and phase spectrum.

8 L3 3 1

8 L3 3 1

5. a) Find the DTFT of the following signals

i. $x[n] = \left(\frac{1}{2}\right)^n u[n-4]$

ii. $x[n] = 2^n u[-n]$

8 L3 4 1

- b) Evaluate the FT for the following signals

i. $x(t) = e^{-3t} u(t-1)$

ii. $x(t) = te^{-2t} u(t)$

8 L3 4 1

6. a) Obtain the frequency response and the impulse response of the system having the output $y[n]$ for the input $x[n]$ as given below:

$$x[n] = \left(\frac{1}{2}\right)^n u[n]; \quad y[n] = \frac{1}{4}\left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{4}\right)^n u[n]$$

8 L3 4 1

- b) Find the inverse DTFT of the following

i. $X(e^{j\Omega}) = 1 + 2\cos(\Omega) + 3\cos(2\Omega)$

ii. $Y(e^{j\Omega}) = j[3 + 4\cos(\Omega) + 2\cos(2\Omega)]\sin(\Omega)$

8 L3 4 1

Unit – III

7. a) Find the Z-Transform of the following

i. $x[n] = 3^n u[n] - 2^n u[-n-1]$

ii. $y[n] = \left(\frac{1}{2}\right)^n \sin\left(\frac{\pi}{2}n\right) u[n]$

8 L3 5 1

- b) Find Inverse Z-Transform of $X(z) = \frac{z(z+1)}{(z-1)^2(z-\frac{1}{2})}$ ROC: $|z| > 1$

8 L3 5 1

8. a) Find the transfer function and the impulse response for the causal LTI system described by the difference equation

$$y[n] = y[n-1] + y[n-2] + x[n-1]$$

8 L3 5 1

- b) Consider a signal $y[n]$ which is related to the signals $x_1[n]$ and $x_2[n]$

by $y[n] = x_1[n+3] * x_2[-n+1]$, where $x_1[n] = \left(\frac{1}{2}\right)^n u[n]$ and

$x_2[n] = \left(\frac{1}{3}\right)^n u[n]$, use the properties of Z-Transform to determine the Z-Transform of $y[n]$.

8 L3 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
